SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
 <input type="text" name="name">
 <input type="email" name="email">
 <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.

5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
 <head>
 <title>Company</title>
 </head>
 <body>
 <div>
 <h1>Welcome
 <p>About us
 </div>
 </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
 <input type="text" name="username">
 <input type="password" name="password">
 <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

*I **<u>G.Mohammad Shareef / 192311288</u>** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in* *disciplinary action.*

Signature of the student: ***<u>G.Mohammad Shareef</u>***

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Answers

1)Online Symposium Portal

1. Identify missing HTML5 semantic elements

- <html>
- <head> o <title>
- < main >
- <section>
- < label> for form fields

2. Standards-compliant HTML

```
CTYPEhtml>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Symposium Registration</title>
</head>
<body>

<section>
<h2>Registration Form</h2>

<form action="/register" method="POST">
  <label for="name">Name</label>
  <input type="text" id="name" name="name" required>

  <label for="email">Email</label>
  <input type="email" id="email" name="email" required>

  <button type="submit">Submit</button>
</form>

</section>
</main>
</body>
```

3. HTTP Request

- Method: POST • URL: /
register • Data sent:

name=John

email=iohn@example.com

4. HTTP Response cycle

1. Browser sends POST request.
2. Server processes the data.
3. Server returns a response (e.g., 200 OK or 302 Redirect).
4. Browser displays confirmation page.

5. Usability & Accessibility Improvements

Use < label> elements.
Add required validation.
Use semantic tags (<main>, <section>) .• Provide clear button text.
Ensure keyboard accessibility.

2. Cross-Browser Compatibility Issue

1. Improper Nesting & Missing Closing Tags

- <h1> not closed. • <n>not
closed.
- Missing <!DOCTYPE html>.

2. Browser Interpretation

- Browsers may automatically close missing tags. o
Different browsers may display different layouts.

3. Missing Semantic Elements

< head e r >
<main>
<section>

4. Correct W3C-Compliant HTML

```
CTYPE html>
<html lang="en">

<meta charset="UTF-8">
<title>Company</title>
</head>



<header>

<hl ·Welcome<>

</header>


<main>
<section>
<p>About us</p>
</section>
</main>

</body>
</html>
```

5. Cross-Browser Compatibility Techniques

Use valid HTML5.
Test on multiple browsers.
Use CSS Reset/Normalize.

Validate HTML and CSS. ● Use responsive design.

3. Form Submission Failure

1. Issues in Form Configuration

Uses GET instead of POST.
Button type is button " instead of submit".

2. Why HTTP Request is Not Triggered

● type= " button does not submit the form. ● Only type="
submit ' sends the request.

3. Role of GET Method

Sends data in the URL.
Suitable for search forms.
Not secure for passwords.

4. Correct Implementation

```
<form action="/submit" method="POST">

<label>Username</label>
<input type="text" name="username" required>

<label>Password</label>
<input type="password" name="password" required>

<button type="submit">Login</button>

</form>
```

5. Secure Data Transmission Improvements

Use POST method.

Use HTTPS.
Validate input on client and server.
Hash passwords before storing.
Enable authentication and session management.